Input graph $G = (V, E, X)$ Support labels Y_S

Support labels only; no query labels

Frozen backbone + task tokens

Transductive few-shot classification

M1 Semantic gating

Propagate + center

$$R = \widehat{A}^2 X$$

Prototype refinement

Support + soft prototypes

$$c^{(j+1)} = 0.5\, o + 0.5\, p^{(j)}$$

Sharpen + propagate

$$\Pi_T = \mathrm{softmax}(0.2B/T)$$

$$Q = \widehat{A}(\Pi_T - \overline{\Pi}_T)$$

Temperature: 0.2 → 0.1

$Q \in \mathbb{R}^{N \times C}$

Fixed teacher and gate

GNN Frozen backbone

Frozen GCN layer 1

$$U = \mathrm{GCN}_1(X, E)$$

Task-token injection

$$\Delta U = 10\,[Q\tanh(P)] \odot s_U$$

$$H = \mathrm{LeakyReLU}(U + \Delta U)$$

Frozen GCN layer 2

$$Z_P = \mathrm{GCN}_2(H, E)$$

M2 Direction–scale matching

$$\mathcal{L} = \mathcal{L}_{\mathrm{LOO}} + \lambda\, \mathrm{mean}(P \odot P)$$

$$D_t = L_t^{-1/4} G_t R_t^{-1/4}$$

$$s_t = \frac{\|A_t^{\mathrm{Adam}}\|_F}{\|D_t\|_F}$$

$$P_{t+1} = P_t - \eta\, s_t D_t$$

P only · zero init. · 100 steps

M3 Clamped propagation

Center embeddings

$$\overline{Z} = Z_P - \mathrm{mean}_V(Z_P)$$

Prototypes

F_0

Support labels

50-NN graph

W 50-NN

Clamped diffusion

$$F^{(t+1)} = 0.1F_0 + 0.9WF^{(t)}$$

$$F_S^{(t+1)} = Y_S$$

Clamp from initialization · 20 steps

Predict $\hat{y}_i = \arg\max_c F_{ic}^{(20)}$

Forward flow · · · Update

P: trainable tokens

Readout only; no backpropagation